

Chemical Vapour Deposition of Bismuth Oxyselenide ($\text{Bi}_2\text{O}_2\text{Se}$) Thin Films on Silicon Substrates

Experimental Methodology

1. Introduction & Material Properties:

This document describes the chemical vapour deposition (CVD) procedure employed to synthesise bismuth oxyselenide ($\text{Bi}_2\text{O}_2\text{Se}$) thin films on silicon dioxide (SiO_2/Si) substrates. The process involves controlled vapour-phase transport of bismuth oxide (Bi_2O_3) and bismuth selenide (Bi_2Se_3) precursors within a tube furnace under a regulated nitrogen atmosphere.

Motivation: Moore's Law limitations in silicon-based transistors necessitate alternative semiconductors. Bismuth Oxyselenide ($\text{Bi}_2\text{O}_2\text{Se}$) offers high electron mobility, air stability, and ultra-thin layered structure ideal for next-generation electronics.

$\text{Bi}_2\text{O}_2\text{Se}$ Properties: Multilayer oxychalcogenide with small indirect bandgap (~ 0.8 eV), high electron mobility, low thermal conductivity, and layered structure of $[\text{Bi}_2\text{O}_2]^+$ and Se^- layers held by weak electrostatic forces. Electronic properties are thickness-dependent due to quantum confinement, enabling bandgap engineering.

CVD Advantage: Provides excellent control over thickness, composition, and consistency across large substrate areas. Compatible with semiconductor manufacturing. Oxygen-assisted powder-precursor CVD is the most widely adopted method.

2. Experimental Methodology:

2.1 Materials and Reagents

- Bismuth (III) oxide (Bi_2O_3): 16 mg (source precursor)
- Bismuth selenide (Bi_2Se_3): 48 mg (selenium source; 1:4 or 1:3 molar ratio with Bi_2O_3)
- SiO_2 substrate chips (growth substrate)
- Furnace: Horizontal Quartz tube (precursor holder)
- Acetone and isopropyl alcohol (IPA) for cleaning
- Carrier gas: Nitrogen (N_2)
- Argon or Nitrogen gas (for substrate cleaning).
- Base Vacuum Pressure: 0.018-0.023 mbar.

2.2 CVD Process Steps

2.2.1 **Cleaning:** Substrate/boat cleaned with acetone (5 min) + IPA (5 min), dried with argon

2.2.2 **Loading:** Precursors loaded in boat - substrate positioned 7.5 cm downstream from Bi₂O₃ source

2.2.3 **Vacuum Purging:** Evacuate to 0.018 – 0.023 mbar, purge with N₂ at 40 sccm

2.2.4 **Growth:** Ramp 30°C to 820°C (19 min) at 40 sccm N₂, hold at 820°C for ≥1 hour

2.2.5 **Cooling:** Natural cool-down (~2 hours) at 100 sccm N₂

3. Experimental Procedure:

3.1 Substrate and Equipment Cleaning

- Clean the quartz boat and SiO₂ substrate sequentially with acetone and IPA (5 minutes each solvent)
- Dry and purge the quartz boat and substrate with argon or another inert gas to remove residual solvent and surface contaminants.

3.2 Precursor Preparation and Loading

- Weigh Bi₂O₃ (16 mg) and Bi₂Se₃ (48 mg) to achieve a 1:3 molar ratio (or 1:4 ratio as required by experimental design).
- Either mix both precursors in the quartz boat or place them separately at a distance of 4 cm apart within the boat.
- Position the SiO₂ substrate chip 7.5 cm downstream from the Bi₂O₃ source (i.e., 6–7 cm from the Bi₂O₃ zone).
- Load the quartz boat into the tube furnace and seal the system.

3.3 Vacuum and Gas Purging

- Evacuate the tube furnace using the vacuum pump. Allow the system to degas for approximately 5 minutes at the start, or at least till the pressure is steady.
- Monitor the base vacuum pressure; target ~0.018–0.023 mbar before proceeding.
- Close the pump valves once stable vacuum is achieved.
- Purge the system with N₂ gas at 40 sccm to remove residual oxygen and moisture. Confirm pressure stabilises (pressure-stable indicator).

3.4 CVD Growth Run

- Initiate the furnace heating ramp per the programmed profile.
- Maintain N₂ carrier gas flow throughout the growth run.
- Record system pressure at key intervals during the run (e.g., after 20 min and after 1 hour).
- Upon completion, allow the furnace to cool naturally under continued N₂ flow before unloading the sample (for cooling maintain the gas flow at 100sccm).
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4. Furnace Temperature Profile

Figure 1 illustrates the temperature ramp profile used during the CVD growth runs.

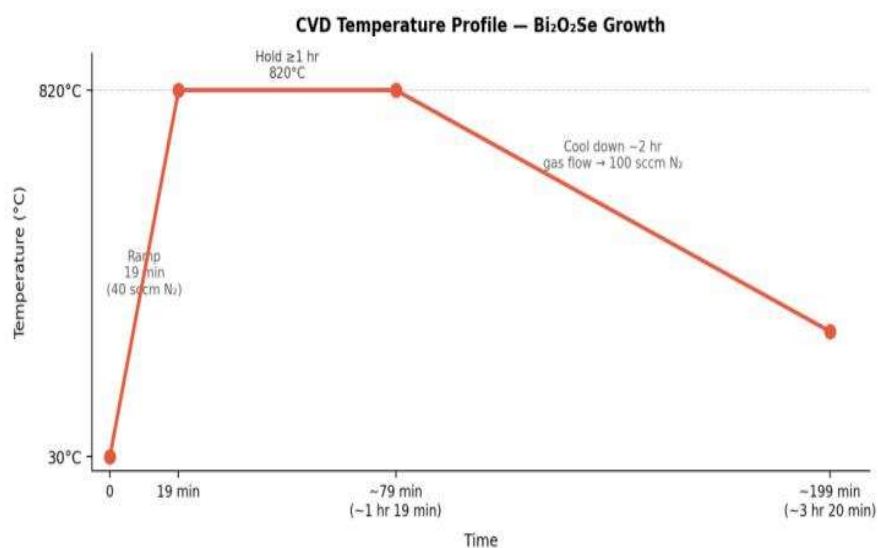


Figure 1. CVD temperature profile: ramp from 30°C to 820°C (19 min, 40 sccm N₂), hold at 820°C (≥1 hr), then cool-down (~2 hr, 100 sccm N₂).

4.1 Temperature Profile

Phase	Temperature	Duration	N ₂ Flow
Ramp up	30°C - 820°C	19 min	40 sccm
Hold up	820°C	≥1 hour	40 sccm
Cool down	820°C - Room temp	~ 2 hours	100 sccm

5. Raman Spectroscopy Analysis:

5.1 Observed Peaks & Interpretation

Peak (cm ⁻¹)	Assignment	Directions	Status	Significance
60	E1 g Bi-O vibrations	xy- plane	Observed	Confirms partial oxidation; Bi-O bonding in xy plane present
160	Bi-O vibrations	z- direction (out of plane)	Missing	CRITICAL - Should be confirmatory for phase-pure Bi ₂ O ₂ Se
95	Unknown	----	Observed	Requires identification; possible secondary/intermediate phase

5.2 Key findings:

Peak at 60 cm⁻¹ (E1g mode): Confirms Bi-O bonding in xy plane, indicating successful oxidation initiation. However, this peak alone is NOT confirmatory for Bi₂O₂Se as it can appear in other bismuth oxide phases.

Missing Peak at 160 cm⁻¹ (CRITICAL): This out-of-plane Bi-O stretching mode is the definitive signature of phase- pure Bi₂O₂Se. Its absence indicates incomplete oxidation and incomplete phase formation. Bi-O bonds are NOT forming in the z direction (perpendicular to layers), which is essential for the complete Bi₂O₂Se structure.

Peak at 95 cm⁻¹ (Unknown): Possible origins include (1) secondary bismuth oxide phase, (2) defect-related vibrational mode, (3) intermediate phase during Bi₂Se₃ to Bi₂O₂Se transformation, or (4) combination mode. Requires further investigation.

6. Analysis & Implications:

Current Material State: Films show partial oxidation with incomplete phase formation. Bi-O bonding occurs in xy plane but lacks critical z-direction bonding. Material is in intermediate oxidation state, not phase-pure Bi₂O₂Se.

Process Status: CVD setup is properly designed and functional. The temperature profile is appropriate. However, current conditions are insufficient for complete oxidation and phase-pure Bi₂O₂Se formation.

7. Conclusions:

CVD synthesis of $\text{Bi}_2\text{O}_2\text{Se}$ has been successfully initiated, with Raman spectroscopy confirming partial oxidation through the presence of the 60 cm^{-1} E1g peak. However, the critical absence of the 160 cm^{-1} confirmatory peak indicates incomplete phase formation. The material currently exists in an intermediate oxidation state with in-plane Bi-O bonding but lacking the essential out-of-plane bonding characteristic of phase-pure $\text{Bi}_2\text{O}_2\text{Se}$.

Systematic optimization of CVD parameters—particularly oxygen content, temperature, and hold time is necessary to achieve phase-pure $\text{Bi}_2\text{O}_2\text{Se}$. Once optimal conditions are established through the recommended parametric studies and complementary characterization, the resulting films will be suitable for integration into high-performance electronic and optoelectronic devices, realizing the significant potential of this promising material system.

8. Challenges & Solutions:

Some of the biggest problems are losing selenium, controlling oxygen, and creating defects. These are dealt with by circumstances high in selenium, regulated oxygen flow, optimum temperature gradients, and delayed cooling in an inert atmosphere.

9. Applications:

$\text{Bi}_2\text{O}_2\text{Se}$ thin films are good for high-mobility field-effect transistors, low-power logic devices, and infrared/visible photodetectors because they are stable, have high mobility, and can be made in large quantities.

10. References:

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